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Inverse design and experimental verification of an acoustic sink based on machine learning

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ABSTRACT

Recently, advances in artificial intelligence research, in particular "machine learning", made it much easier to predict the structural parameters for a specific desired acoustic performance. In this paper, we build a acoustic sink model by using transfer matrix method, and establish a database of 79,730 lines in the frequency domain of 0–5 kHz. The proposed CNN model, which consists of two building blocks (encoder and decoder), is expected to predict the corresponding geometric parameters of the selected (expected) absorption curve. Geometric parameters include diameter *a* of the neck structure, diameter *b* of the cavity, neck length *c*, and thickness *e* of porous material. Through the comparison of four groups of prediction curves, target curves and test curves, it can be concluded that accuracy of prediction is high in the sound absorption area. This study verifies the feasibility of machine learning method in the inversion design of acoustic functional devices, which is suitable for performance structure inversion prediction of complex acoustic structures, and has potential application in shortening the design cycle of acoustic products.

1. Introduction

The design of acoustic functional materials or structures is an active, rapidly developing field of research in recent years [1–4]. Novel acoustic functional structures have exciting potential applications for reduction of vibration and noise [5–9], acoustic filtering [10], and topological acoustics [11–13]. In order to continuously optimize the sound absorption performance, most of the previous studies focused on the use of optimization algorithm [14–18]. Although the goal is to improve the average absorption coefficient, the calculation time is heavily dependent on the complexity of the analytical model, and it is still unable to reverse the structure design according to the acoustic performance.

With the rapid development of artificial intelligence, machine learning in particular has become a powerful tool that is now widely used in data mining, computer vision, and language processing [19,20]. In recent years, machine learning has been used to design even new materials [21–23]. This is a paradigm change for material design, which made it possible to be less reliant on experience and theoretical knowledge. The use of machine learning in this field also helped to accumulate experience and summarize rules for the design of new structures, and it became possible to find solutions for many challenging real-life requirements. In design of nano-photonics [21], electromagnetic meta-surface [22], and mechanical metamaterials [23], machine learning already showcased the benefits of the new "on demand" design-approach. For a sound insulation structure [24], the transmission curve, which was predicted using machine learning, was practically identical with the target curve. This high accuracy is a strong argument for the new forward-design approach of acoustic structures. However, the application of machine learning to resonant acoustic absorption structures has not yet been reported. In this paper, an acoustic-sink, which are composed of a Helmholtz resonator and a porous material, is regarded as the research object that needs to be predicted. By training the model using a database of samples (consisting of data of structure sizes and sound absorption curves), and a convolutional neural network (CNN), the dimensions of a new structure (which is not part of the database) can be predicted for a desired (target) sound characteristic (sound-absorption curve). Finally, by comparing the target absorption curve with test data, the feasibility of machine learning for the design of composite absorbers was confirmed. This result advances the design of new acoustic functional devices substantially.

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2. Acoustic-sink model and forecast database

The acoustic sink structure, which is proposed in this paper, is shown in Fig. 1(a) and (b). Here, p_i and p_r denote the incident and reflected sound pressures, respectively. Parameter d_t = 99 [mm] is the spatial diameter of the acoustic energy radiated from the acoustic sink. Variable a is the diameter of the neck structure, variable b represents the diameter of the cavity, variable c is the neck length, variable e is the thickness of porous material, and variable l is the cavity length. If the aperture of the neck is small, the air in the acoustic sink will have a strong thermal viscosity effect, when it propagates within it. Using a thermal viscosity loss model [25], the thickness of a viscous boundary layer can be expressed as:

$$d_v = \sqrt{2\mu/\rho_0\omega} \tag{1}$$

Here, the aerodynamic viscosity μ = 1.568 × 10^{-5}m^2/s, the air density ρ_0 = 1.29 kg/m^3, and the working frequency $\omega = 2\pi f$.

The normalized equivalent wave number k_i, and the normal acoustic impedance Z_i in the neck and cavity are [25]:

$$k_i = \frac{\omega}{c_0}\left(1+\frac{\beta}{s_i}(1+(\gamma-1)/\chi)\right) \tag{2}$$

$$Z_i = \frac{\rho_0 c_0}{S_i}\left(1+\frac{\beta}{s_i}(1-(\gamma-1)/\chi)\right) \tag{3}$$

Here, $i = n$ or c represent the neck or cavity of the acoustic sink, and the velocity in air c_0 = 343 m/s. Furthermore, $s_i = b/2d_v$, $\beta = (1+i)/\sqrt{2}$, $\chi = \sqrt{\mathrm{Pr}}$, the specific heat rate in air γ = 1.4, and the Prandtl number at one atmospheric pressure Pr = 0.707. For homogeneous porous materials, the Johnson Champous Allard (JCA) model is used, and the equivalent density of porous materials can be written as [26–28]:

$$\rho_p(\omega) = \frac{\rho_0\alpha_\infty}{\varphi}\left(1-i\frac{\omega_c}{\omega}F(\omega)\right) \tag{3-1}$$

$$F(\omega) = \sqrt{1-\frac{1}{i}\mu\rho_0\omega\left(\frac{2\alpha_\infty}{\sigma\varphi\Lambda}\right)^2} \tag{3-2}$$

The equivalent bulk modulus of porous materials is [26–28]:

$$K_p(\omega) = \frac{\gamma P_0}{\varphi\left(\gamma-(\gamma-1)\left(1-i\frac{\omega_c^*}{\mathrm{P_r}\omega}G(P_r\omega)\right)^{-1}\right)} \tag{4-1}$$

$$\omega_c = \frac{\sigma\varphi}{\rho_0\alpha_\infty},\ \omega_c^* = \frac{\sigma^*\varphi}{\rho_0\alpha_\infty},\ \sigma^* = \frac{8\alpha_\infty\mu}{\varphi\Lambda'} \tag{4-2}$$

$$G(\mathrm{P}_r,\omega) = \sqrt{1-\frac{1}{i}\eta\rho_0\mathrm{P_r}\omega\left(\frac{2\alpha_\infty}{\sigma^*\varphi\Lambda'}\right)^2} \tag{4-3}$$

Here, P_0 = 1.013 × 10^5 Pa and the tortuosity α_∞, porosity φ, the thermal characteristic length Λ', the viscous characteristic lengths Λ, the flow resistance σ are provided by a BOSCH Acoustic Technology Detector – see Table 1**.** Porous material was provided by the Shenzhen Jiexin rubber sponge company. In order to verify the accuracy of parameter setting in Table 1, we conducted sound absorption test on pure porous material, and compared it with the theoretical sound absorption coefficient based on JCA model. The results are shown in Fig. 1. In the range of 0–5 kHz, the two curves are basically identical, but there is a certain error, which is due to the temperature deviation between the test environment and the theoretical model, which leads to the inconsistency of thermal viscosity loss. In addition, there will be some noise interference and leakage in the test, which will affect the accuracy of the test results.

When the sound wave enters the acoustic sink unit vertically, it passes through the neck, cavity, and porous material. We use transfer matrix method to obtain the sound absorption coefficient α. Assuming that the sound pressure incident from the left end of the structure is p_1, the normal sound velocity is v_1, the sound velocity at the end of the structure is p_2, and the normal sound velocity is v_2, then the sound wave propagation equation is [29,30]:

$$\begin{pmatrix} p_1 \\ v_1 \end{pmatrix} = \mathrm{T}_n\mathrm{T}_{\Delta l}\mathrm{T}_c\mathrm{T}_p\begin{pmatrix} p_2 \\ v_2 \end{pmatrix} \tag{5}$$

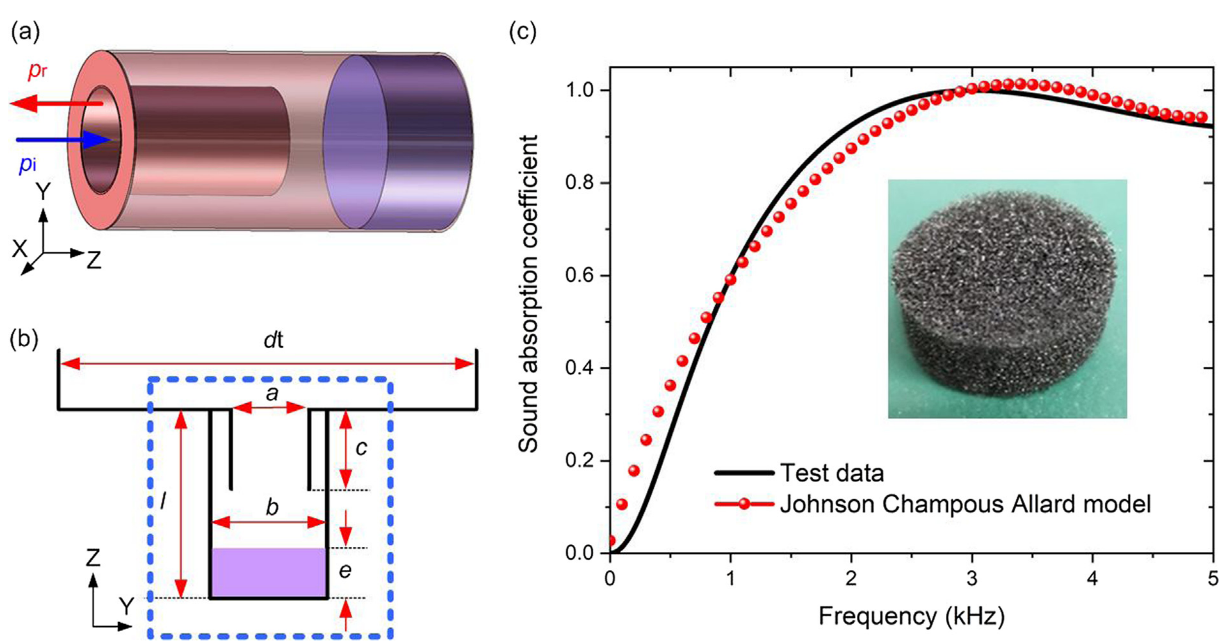

Fig.1. The sketch map of an acoustic sink. Basic unit (a), cross-section (b), and Theoretical predicted values of five parameters in JCA model and test dta.

Table 1
Five acoustic parameters for the JCA model.

Acoustic parameters	α_∞,	φ	Λ'	Λ	σ
Value	1.15	0.93	127 [μm]	62 [μm]	29,370 [Pa·s]

Here, T_n is the transfer matrix of the neck structure, $T_{\Delta l}$ is the end length correction matrix when the sound wave propagates from the neck to the cavity and from the neck to the outer space, T_c is the transfer matrix of the cavity structure, and T_p is the transfer matrix of the homogeneous porous material.

Here, the transfer matrix of neck and cavity is [29,30]:

$$T_n = \begin{pmatrix} \cos(k_n l_n) & iZ_n \sin(k_n l_n) \\ \frac{i}{Z_n}\sin(k_n l_n) & \cos(k_n l_n) \end{pmatrix} \tag{6}$$

$$T_c = \begin{pmatrix} \cos(k_c l_c) & iZ_c \sin(k_c l_c) \\ \frac{i}{Z_c}\sin(k_c l_c) & \cos(k_c l_c) \end{pmatrix} \tag{7}$$

When the sound wave propagates from the neck to the cavity and from the neck to the outer space, the normal sound pressure at the abrupt section is discontinuous, and the correction matrix of the end length is [29,30]:

$$T_{\Delta l} = \begin{pmatrix} 1 & iZ_n k_n \Delta l \\ 0 & 1 \end{pmatrix} \tag{8}$$

Here, Δl is the end correction length [29,30]. When the sound wave propagates from the neck to the cavity or from the neck to the outer space, there is the acoustic impedance of the end radiation caused by the discontinuous interface. At this time, it is necessary to correct the length of the end of the neck, and the correction lengths are [3,7–9]:

$$\Delta l_1^{corr} = 0.82\left[1 - 1.35\left(\frac{r_n}{r_c}\right) + 0.35\left(\frac{r_n}{r_c}\right)^3\right] r_n \tag{9}$$

$$\Delta l_2^{corr} = 0.82\left[1 - 0.235\left(\frac{r_n}{r_t}\right) - 1.32\left(\frac{r_n}{r_t}\right)^2 + 1.54\left(\frac{r_n}{r_t}\right)^3 - 0.86\left(\frac{r_n}{r_t}\right)^4\right] r_n \tag{10}$$

Where, $r_n = a/2$, $r_c = b/2$ and $r_t = b/2$ are neck radius, cavity radius and the whole structure radius respectively. The correction term Δl_1^{corr} and Δl_2^{corr} are caused by the discontinuity of the normal pressure at the abrupt section when the sound wave propagates from the neck to the cavity and from the neck to the outer space. So the total correction length is $\Delta l = \Delta l_1^{corr} + \Delta l_2^{corr}$.

The transfer matrix of homogeneous porous material is:

$$T_p = \begin{pmatrix} \cos(k_p l_p) & iZ_p \sin(k_p l_p) \\ \frac{i}{Z_p}\sin(k_p l_p) & \cos(k_p l_p) \end{pmatrix} \tag{11}$$

Here, k_n, k_c, k_p are the equivalent wave numbers of neck, cavity and porous material. l_n, l_c, l_p are the length of neck and cavity and the thickness of porous material. Z_n, Z_c, Z_p are the normal impedance of the neck, cavity and porous material. $Z_p = Z_{p0}/S_c$, Z_{p0} is the normal acoustic impedance of porous materials. Combined with Fig. 1b, $l_c = c$, $l_p = e$, $l_n = l\text{-}c\text{-}e$. The equivalent characteristic impedance of homogeneous porous material Z_p is

$$Z_p = \sqrt{\rho_p(\omega)K_p(\omega)} = \rho_p(\omega)c_p \tag{12}$$

The equivalent sound velocity c_p and the equivalent wave number k_p of porous materials are:

$$c_p = \sqrt{\frac{K_p(\omega)}{\rho_p(\omega)}},\ k_p = \frac{\omega}{c_p} \tag{13}$$

Since the cavity at the end of the whole structure is closed, v_2=0. There are simultaneous equations (5)–(11) to cancel the p_2 in the equation, then the normalized normal acoustic impedance:

$$Z_{as} = \frac{p_1}{v_1} = i\frac{Z_n}{Z_t}\cdot\frac{a}{b} \tag{10-1}$$

$$a = 1 - \tfrac{Z_c}{Z_p}\tan(k_c l_c)\tan(k_p l_p) - \tfrac{Z_n}{Z_c}\tan(k_n l_n)\tan(k_c l_c) - \tfrac{Z_n}{Z_p}\tan(k_n l_n)\tan(k_p l_p) - k_n\Delta l Z_n\left(\frac{\tan(k_c l_c)}{Z_c} + \frac{\tan(k_p l_p)}{Z_p}\right) \tag{10-2}$$

$$b = \tan(k_n l_n) - \tfrac{Z_c}{Z_p}\tan(k_c l_c)\tan(k_p l_p)\tan(k_n l_n) + \tfrac{Z_n}{Z_c}\tan(k_c l_c) + \tfrac{Z_n}{Z_p}\tan(k_p l_p) - k_n\Delta l Z_n\left(\frac{\tan(k_c l_c)\tan(k_n l_n)}{Z_c} + \frac{\tan(k_p l_p)\tan(k_n l_n)}{Z_p}\right) \tag{10-3}$$

Here, $Z_t = \frac{\rho_0 c_0}{\pi\cdot(d_t/2)^2}$ is the equivalent acoustic impedance of the tube.

Therefore, the reflection coefficient r and absorption coefficient α of the sound absorption unit are as follows [31]:

$$r = \frac{Z_{as} - 1}{Z_{as} + 1} \tag{11}$$

$$\alpha = 1 - |r|^2 \tag{12}$$

Next, we built a sample database with four sensitive geometric parameters a, c, b, and e, which directly affect the sound absorption α. Machine learning is then used to facilitate the on-demand creation of a specific acoustic characteristic. Sample database plays an important role in training results and prediction accuracy. The more uniform the distribution of data is in the database, the larger is the range of data, and the higher is the accuracy of the prediction. For 1 to 5 kHz, four geometric parameters in each group correspond to a sound absorption curve. This can be considered a set of 5000 dimensional data. Using different combinations of the geometric parameters a, c, b, and e, 79,730 different sound absorption curves can be constructed. In order to describe the distribution of 79,730 sets of line data as much as possible, we selected 233 sets of line data for display, see Fig. 2. There are two clear absorption peaks in each absorption curve within 5 kHz, and the highest absorption peak is between 592 Hz and 1577 Hz. This suggests a very clear resonance absorption effect. At this time, the addition of porous materials of varying thickness alters the loss of acoustic energy in the resonant cavity, and the sound absorption peaks here represent different heights. Therefore, the distribution of the sample database can be as uniform as possible - see the target region (dotted line box) in Fig. 2.

3. Framework and method of deep learning

The main approaches of deep learning include Artificial Neural Networks (ANN), Convolution Neural Network (CNN) and Recurrent Neural Networks (RNN). ANN can also be used for modeling between the performance curve and structure parameters. However, the connection of neurons is dense, which makes the param-

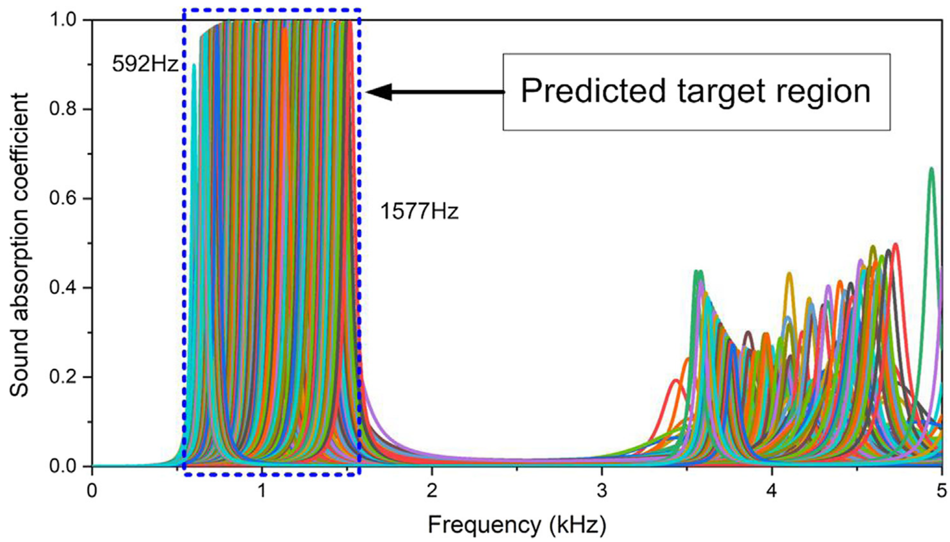

Fig. 2. Preselected 233 representative sound absorption curves of a sample database.

eters of networks increase dramatically [32]. That means the much more data will be need. Therefore, CNN is more efficient than ANN in this work. RNN is used for sequence modeling, such as text [33] and speech processing [34–36]. However, there is no sequence in our data. Therefore, RNN is not suitable for this work. In our work, the performance curve is a one-dimensional vector with 5000 points. One-dimensional CNN can extract the local characteristics from the vector [32]. Therefore, CNN is suitable to address this issue and CNN based encoder-decoder network is adopted in our work. In addition, the CNN can effectively capture the local characteristics thanks to its local connection, which is why we use it to train it and identify the sample database. Due to the one dimensional characteristic of the sound absorption curve, one dimensional convolution (Conv1D) is used to extract the corresponding information.

The sound absorption curve is denoted as α, and the geometric parameter is denoted as y=(y_1, y_2, y_3, y_4)=(a, b, c, e). The proposed CNN model is expected to predict the corresponding geometric parameters for a chosen (desired) sound absorption curve. In this paper, the CNN architecture consists of an encoder-decoder model, which is made up of two building blocks (encoder and a decoder) – see Fig. 3. The encoder f_E extracts low-dimensional features z from the sound absorption curve α, layer by layer, which can be expressed as $z = f_E(\alpha)$. We made the output of the encoder equal to the geometric parameter y, thus, the encoder was able to map the geometric parameter from the required sound absorption curve α, $\widehat{y} = f_E(\alpha)$. Here, $\widehat{y}$ denotes the estimate of the geometric parameter y. The encoder was trained to minimize a loss$L(y, \widehat{y})$ between the geometric parameter y and its estimate $\widehat{y}$. Generally, the mean square error (MSE) was used as loss function. Therefore, the training target of the encoder is as follows:

$$\min L(y,\hat{y}) = \frac{1}{M}\sum_{i=1}^{M}(y_i - \hat{y}_i)^2 \tag{13}$$

Here, M is the number of geometric parameters (a, b, c, e). Using Eq.(13), we can reach the conclusion that the different geometric parameters have the same weight during the process of constantly adjusting the loss. Because of the relationship between each geometric parameter and the corresponding sound-absorption curve is not linear, the sensitivities of the four geometric parameters are different. Therefore, we can not overcome this point only using MSE loss and the weighted MSE. Hence, the decoder is used to evaluate the geometric parameter in a non-linear way, i.e. using the neural network.

The decoder f_D is trained to predict the sound absorption curve α from the input geometric parameters y, $\widehat{x} = f_E(y)$, where $\widehat{x}$ is the estimate of the sound-absorption curve. The decoder, on the other hand, is trained to minimize α MSE loss $L(x, \widehat{x})$ between sound-absorption curve α and its estimate$\widehat{\alpha}$,

$$L(\alpha,\hat{\alpha}) = \frac{1}{N}\sum_{j=1}^{N}(\alpha_j - \hat{\alpha}_j)^2 \; N = 5000 \tag{14}$$

Therefore, the training target for the encoder-decoder model can be expressed as:

$$\min L(y,\hat{y}) + \beta L(\alpha,\hat{\alpha}) = \frac{1}{M}\sum_{i=1}^{M}(y_i - \hat{y}_i)^2 + \beta\frac{1}{N}\sum_{j=1}^{N}(\alpha_j - \hat{\alpha}_j)^2 \tag{15}$$

Here, β is the hyperparameter and set to 0.5 in this paper.

Furthermore, the CNN is used as the encoder. The upper part of Table 2 shows the structure of the encoder. The encoder contains four convolution blocks and a dense layer. In each convolution block, a 1D convolution layer is used to linearly transform the input using a kernel value of three. After convolution, batch normalization (BN) was applied to speed up and prevent overfitting during the training step. Subsequently, a non-linear activation function, such as ReLU, Sigmoid, is used to improve the performance and the learning capability of the network. To further reduce the dimensions of the feature map, a 1D average pooling layer was implemented. The numbers of four convolution layers increases two times. Finally, the feature map, which was obtained from the convolution layer, was flattened into a 1D vector and fed into a dense layer to output the estimate of the geometric parameter $\widehat{y}$.

The structure of the decoder was symmetrical with respect to the encoder. We first transform the estimate of the geometric parameters into a new feature map using a dense layer. Then, the feature map is transformed into an estimated sound-absorption curve using four convolutional blocks. In each block, the feature-map is first expanded, using the interpolate operation, which is an inverse operation of the aforementioned pooling operation. In this paper, the kernel to interpolate is "nearest" if the size is 5. Then, like for the encoder, the convolution layer, followed by

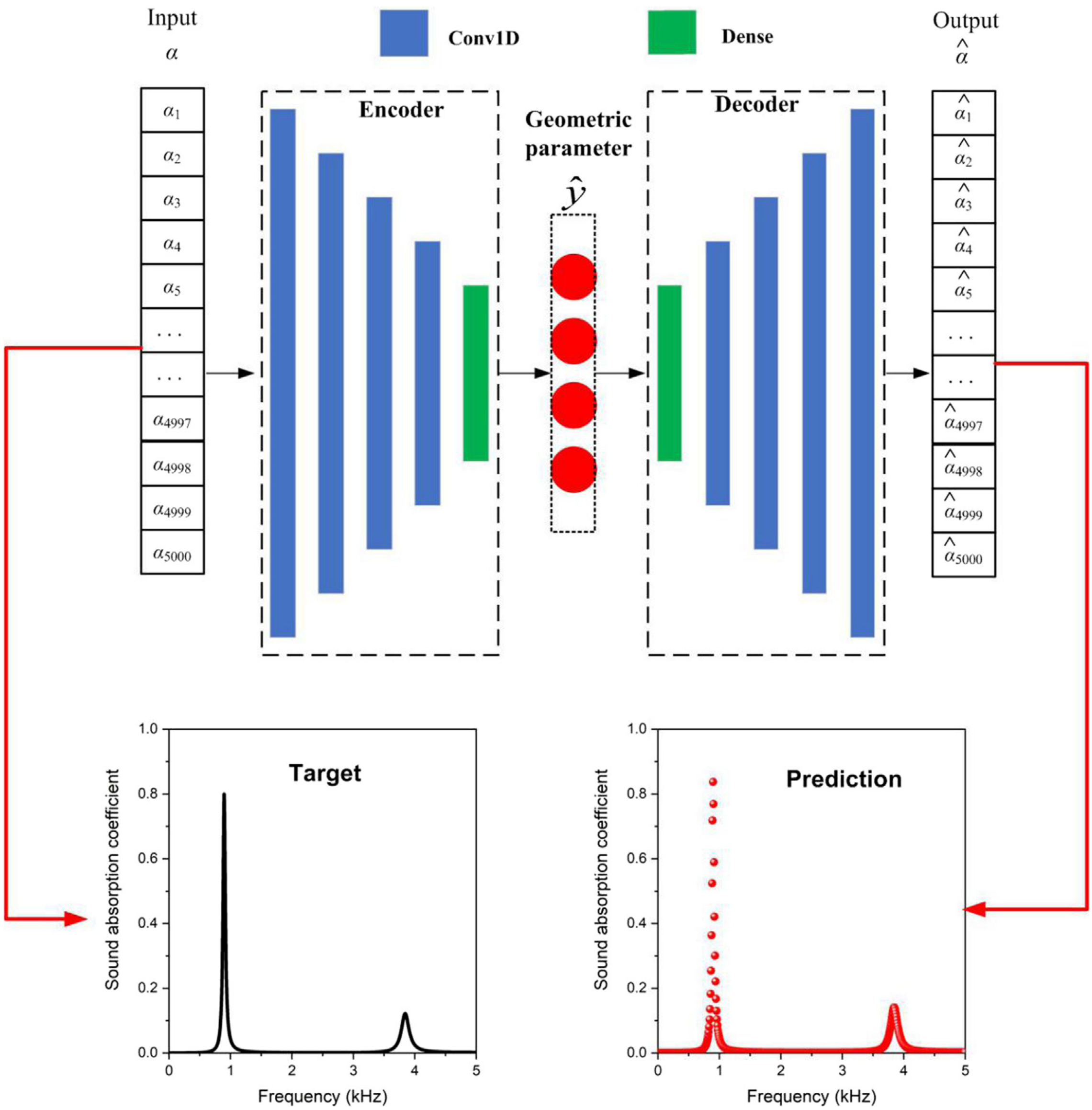

Fig. 3. Schematic of the CNN architecture: encoder and decoder for the encoder-decoder model. Inserted pictures show the target and prediction for the sound-absorption curve.

Table 2
Architecture of the convolutional auto-encoder network. "Conv1D" refers to 1D convolution. "Interpolate" refers to the interpolate operation. "avg-pool1D" refers to the 1D average pooling operation. The numbers in brackets refer to the kernel size of the corresponding operation. The numbers after "@" refer to the number of kernels. BN refers to the batch normalization layer. ReLU and Sigmoid are the activation functions.

Input:	Original sound absorption –5000
Encoder	(3)Conv1D@16-BN-ReLU-(5)avg-pool1D
	(3)Conv1D@32-BN-ReLU-(5)avg-pool1D
	(3)Conv1D@64-BN-ReLU-(5)avg-pool1D
	(3)Conv1D@128-BN-ReLU-(5)avg-pool1D
	Dense-4-Sigmoid
Intermediate output:	geometric parameters(a, b, c, e)
Decoder	Dense-1024-BN
	(5)Interpolate-(3)Conv1D@64-BN-ReLU
	(5)Interpolate-(3)Conv1D@32-BN-ReLU
	(5)Interpolate-(3)Conv1D@16-BN-ReLU
	(5)Interpolate-(3)Conv1D@1-Sigmoid
Output:	Estimated sound-absorption –5000

batch normalization and the non-linear activation function, is used. Finally, the decoder outputs a 5000-dimensional vector, i.e. the estimate for the sound-absorption curve. We trained the encoder-decoder model using the Adam optimization method for 100 epochs. The learning rate was initialized with the value 10^{-3} and decreased by 10% after each epoch. We selected the model with the best performance after the validation set for evaluation. Lastly, the prediction of the sound-absorption curve and the corresponding geometric parameters were obtained.

4. Prediction results and verification

Four groups of different absorption curves are given randomly from the distribution range of the sample database (79730 absorption curves), and four groups of corresponding geometric parameters are predicted by using the convolution codec model (see Table 3). In order to verify the accuracy of the prediction results, the sound absorption coefficient of the predicted structure in Table 3 is calculated by using the transfer matrix method, and then the samples are prepared according to the predicted structure size.

Table 3
Four geometric parameters of the sound-absorption prediction curves.

	a [mm]	b [mm]	c [mm]	e [mm]
Prediction curve 1#	8	13	38	6
Prediction curve 2#	7	16	13	17
Prediction curve 3#	6	17	31	14
Prediction curve 4#	4	17	20	12

The rigid frame of an acoustic sink was then fabricated using a 3D printing system (nanoArch® P130). Photosensitive resin is selected for 3D printing because it can ensure smooth wall to reduce scattering and simulate hard sound field boundary to the greatest extent. The porous materials were selected according to Table 1. Four test samples are shown in Fig. 4. For the test of the sound absorption coefficient, an acoustic impedance tube, two microphones (Brüel and Kjær 4187) and a data-acquisition module (Brüel and Kjær 3560C) were used. The sound absorption test results were obtained using the transfer function method with PULSE software (Brüel and Kjær) according to ISO Standard, and the test process is shown in Fig. 5 [37].

In Fig. 6(a)–(d), using the four geometric parameters in Table 3, the target curves are obtained by equation (12) as the standard to verify the prediction curves, and the test curves can prove the effectiveness of the structure predicted (Table 3) in the actual sound absorption test. In Fig. 6(a) and (c), the coincidence degrees of the target and the prediction curves are high, especially in the area where the absorption peaks are located. But in Fig. 6(b) and (d), the frequencies corresponding to the absorption peaks of the target and the prediction curves move to a certain extent, but the peak values are basically the same. From Fig. 7, we can also see that the prediction accuracy of sample 1# and 3# is high, and the error of prediction curve and target curve is basically controlled in the range of absorption coefficient of plus or minus 0.1. For samples

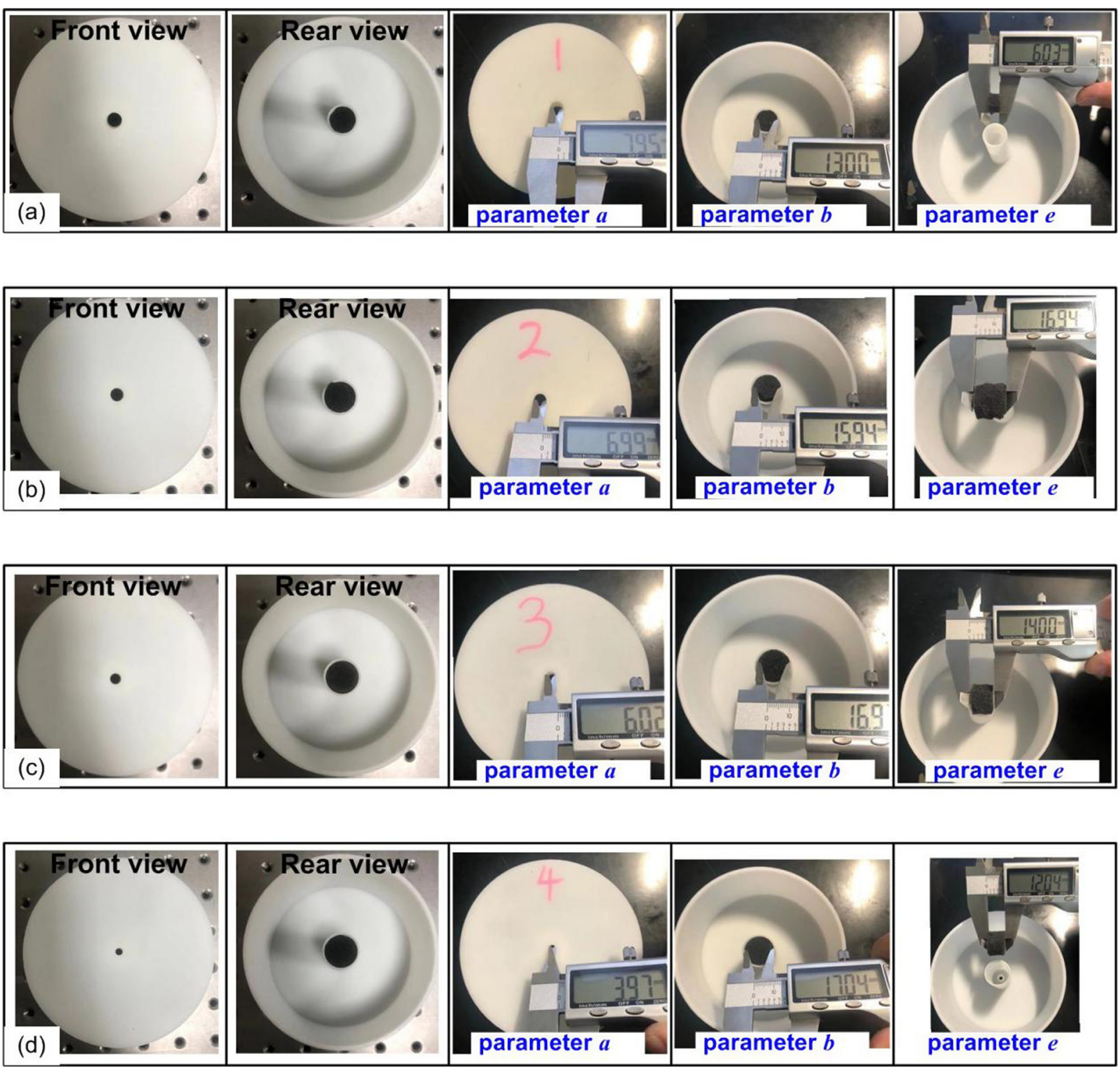

Fig. 4. Test samples 1#-4# corresponding to Table 3.

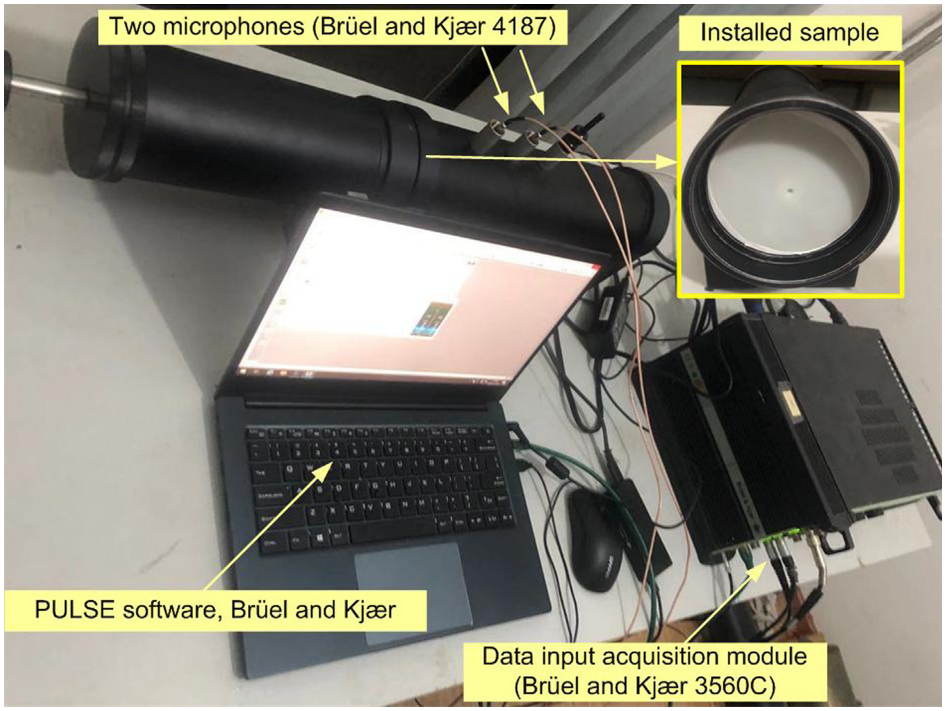

Fig. 5. Sound absorption test using acoustic impedance tube.

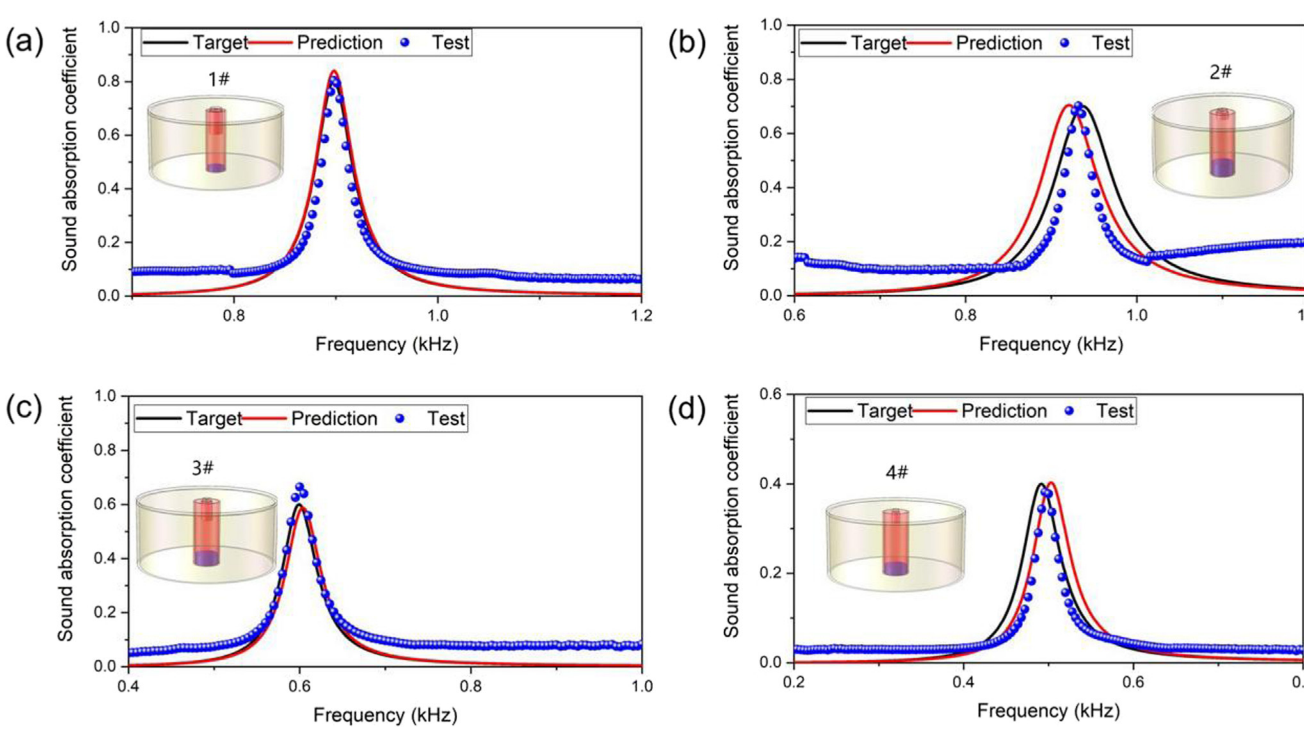

Fig. 6. Four sound-absorption coefficients of the target, prediction, and test, respectively.

2# and 4#, there is a large error between the prediction curve and the target curve due to the influence of the peak frequency shift. The reason for this phenomenon is that although the four groups of random absorption curves are distributed in the sample database, the convolutional codec model represents the corresponding relationship between structure and performance. This kind of "black box" prediction itself has random and inescapable error. In addition, the existence of these prediction errors is also related to the nonlinearity of the structural model itself and the distribution of its sound absorption performance in the frequency domain. In Fig. 6, although the test curves are all higher than the predicted and target curves in the non-absorption peak area, the coincidence degree between these is higher in the absorption peak area. This is because there is a certain gap in the installation error of the sample in the acoustic impedance tube, which may lead to the thermal viscosity loss and improve the sound absorption. In addition, the

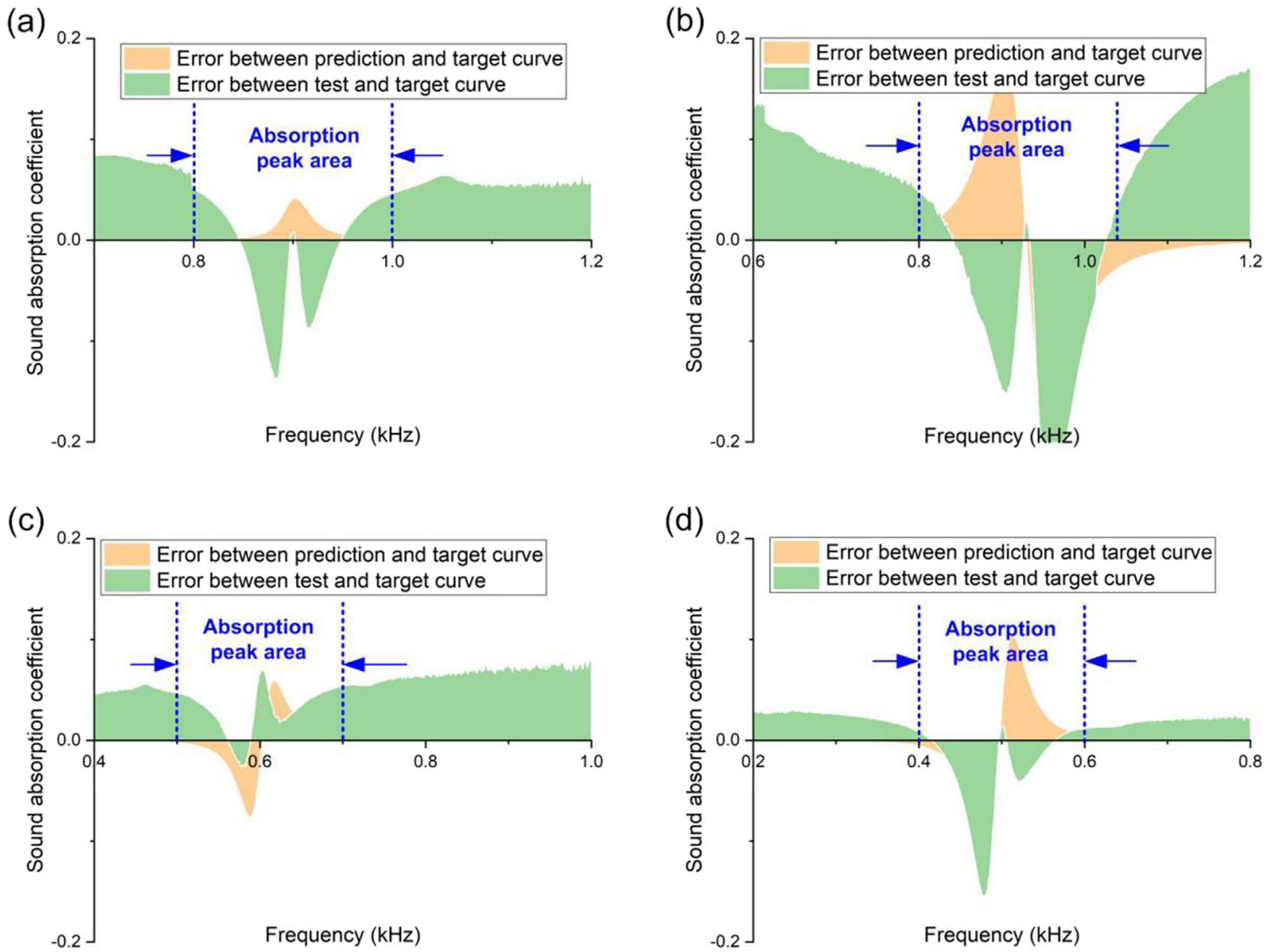

Fig. 7. Errors of prediction curve, test curve and target curve of sound absorption coefficient. (a)–(d) correspond to the sample $1^{\#}$–$4^{\#}$, respectively.

porous material may also improve the sound absorption coefficient in the non-resonant absorption area. The test curves of high absorption area are mainly determined by the structural resonance, so it is basically not affected by the test error. In brief, through the comparison of three types of curves (target, prediction, test), it can be seen that the new convolutional encoder-decoder model can effectively predict the sound absorption curve, as it has been successfully verified by sound absorption tests.

5. Conclusions

In this paper, the applicability of machine learning in the inversion design of sound absorption structure is verified by using a classical resonant-absorption structure, namely sound sink, and convolutional encoder/decoder model. Especially in the absorption frequency domain, the predicted curve is highly coincident with the target curve. This method is different from the previous acoustic product development ideas which only rely on finite element simulation, theoretical equations or engineering experience. The construction of machine learning can make researchers directly obtain the acoustic structure according to the noise reduction requirements of customers, which will undoubtedly greatly shorten the product development cycle. In future work, we will use machine learning as a more general non-analytical acoustic model for prediction, and compare the prediction accuracy of inversion and machine learning.

CRediT authorship contribution statement

Nansha Gao: Investigation, Conceptualization, Funding acquisition. **Mou Wang:** Methodology. **Baozhu Cheng:** Software, Methodology. **Hong Hou:** Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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